

POWERFACTORY

PowerFactory 2024

Technical Reference

HVDC Models in PowerFactory

HVDC LCC Monopole CDC4T

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

Copyright © 2024 DIgSILENT GmbH
All rights reserved. No part of this publication may be reproduced or distributed in any form without written permission of DIgSILENT GmbH.

January 9, 2024
PowerFactory 2024
Revision 1

Contents

1 Introduction **1**

2 *HVDC LCC Monopole CDC4T* Model - Quick Use Instructions **2**

3 Overview of the *HVDC LCC Monopole CDC4T* Model **2**

3.1 Model Usage Guidelines 6

3.1.1 Load Flow Analysis 6

3.1.2 Dynamic RMS-Simulation 6

3.1.3 Dynamic EMT-Simulation 6

4 Model Configuration **7**

4.1 HVDC-LCC element 7

4.1.1 Adjusting Converter Parameters 7

4.2 Dynamic Model Configuration 7

5 References **10**

List of Figures **10**

1 Introduction

This document describes the usage of one High-Voltage Direct Current (HVDC) line commutated converter (LCC) dynamic simulation model, and provided via the *PowerFactory* general template *HVDC LCC Monopole CDC4T* . The dynamic model has been built based on the CDC4T model [1], with a number of additional features.

The model can be used for load flow, RMS- and EMT-Simulation.

2 *HVDC LCC Monopole CDC4T* Model - Quick Use Instructions

The provided HVDC-LCC Dynamic Model consists of one HVDC-LCC element (*ElmHvdclcc*), two AC busbars (one for each station), two common impedance elements (optional elements, may be replaced if necessary) and two shunt compensation devices (*ElmShnt*). The model is delivered as template for the purpose of simple network deployment and ease of use.

A new model (based on the general template) can be added to an active project by following the instructions provided in the *PowerFactory* User Manual, Chapter “Libraries”, section “Templates Library”, subsection “General Templates”.

3 Overview of the *HVDC LCC Monopole CDC4T* Model

The *HVDC LCC Monopole CDC4T* model single line diagram is shown in Figure 3.1.

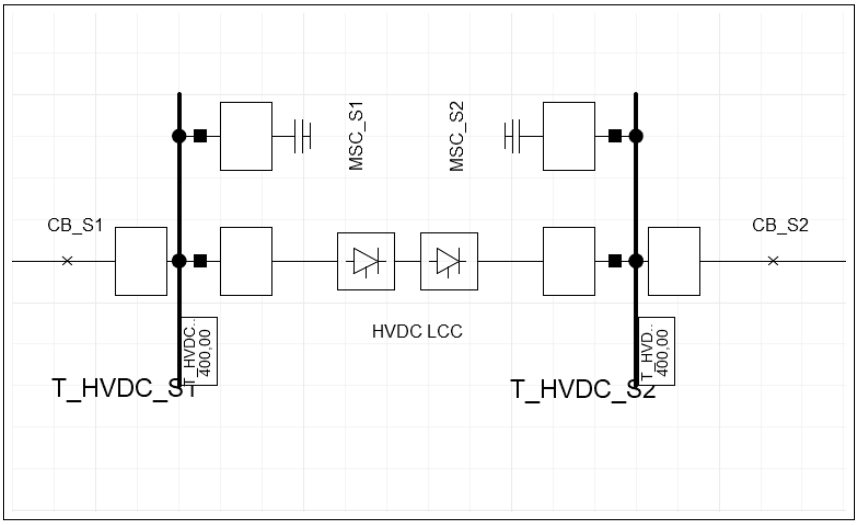


Figure 3.1: HVDC-LCC Single Line Diagram

The dynamic model main composite frame diagram is shown in Figure 3.2.

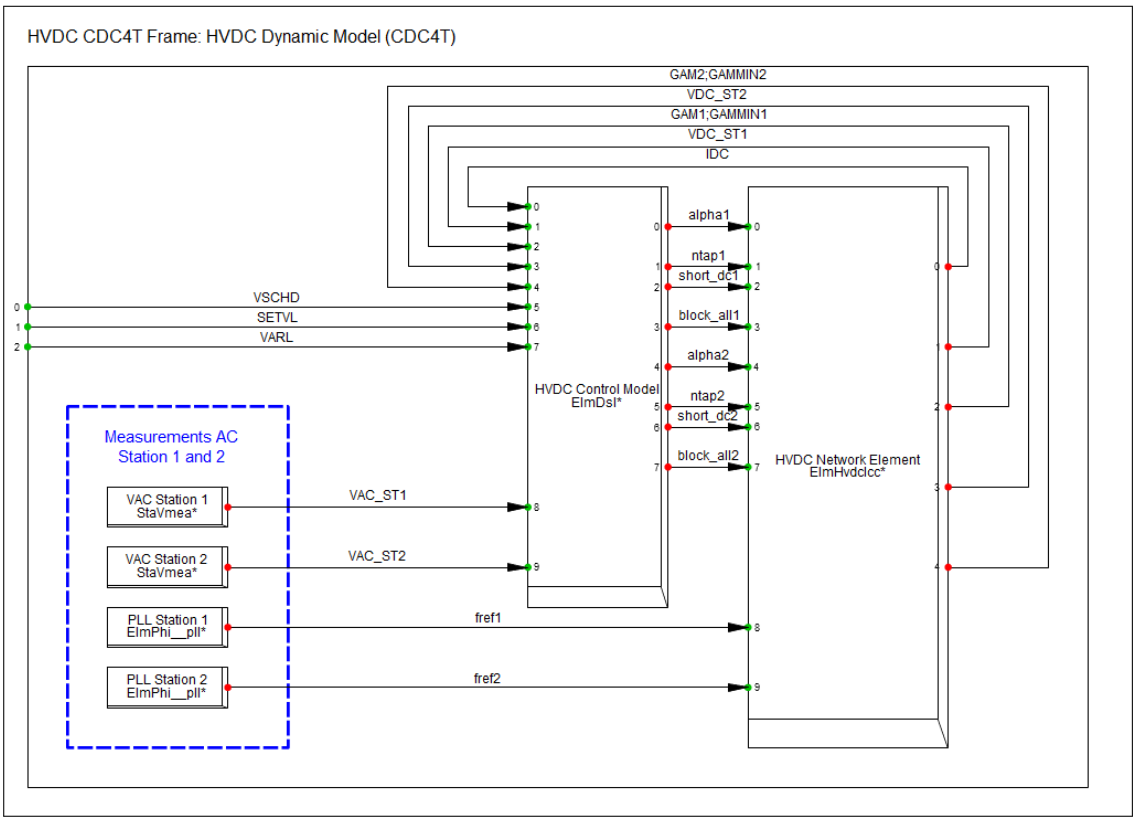


Figure 3.2: HVDC-LCC Composite Frame Diagram

The dynamic model control block diagram is shown in Figure 3.3.

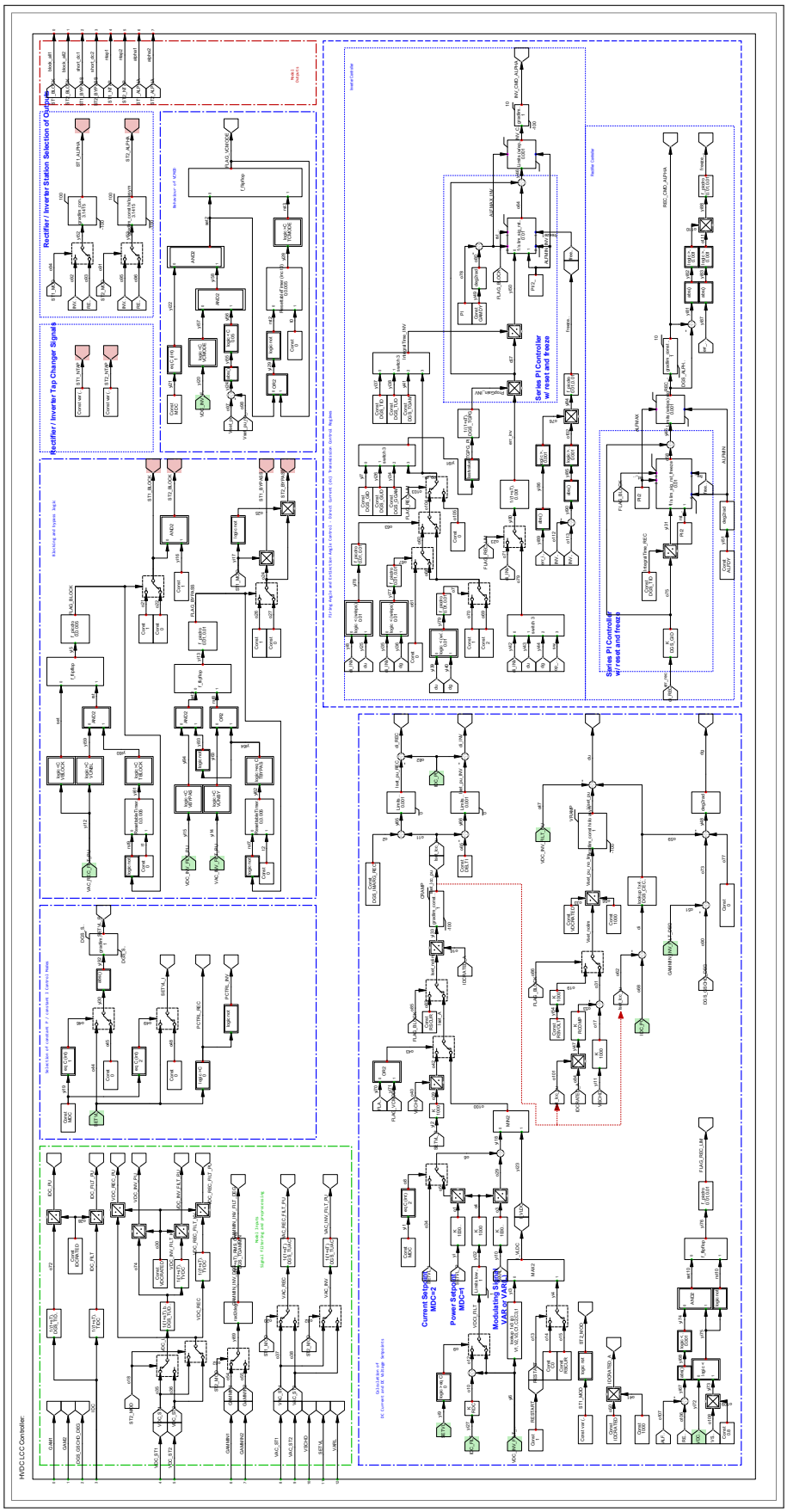


Figure 3.3: HVDC-LCC Control Diagram

The *HVDC LCC Monopole CDC4T* model is delivered as general template. The ratings of the available template (named *HVDC LCC Monopole CDC4T*) are given below:

- Nominal power, Converter Station 1 and 2: 1000 MW
- Nominal voltage, Station 1: 400 kV
- Nominal voltage, Station 2: 400 kV
- Shunt Compensation, Station 1: 450 Mvar.
- Shunt Compensation, Station 2: 450 Mvar.

Note: The above configuration can be customised according to the user requirements by following the instructions in this User Manual.

Due to the fact that shunt compensation equipment is installed to the AC bus of each station, the commutating voltage is located at the respective AC bus (hence, the upstream grid impedance does not play a role to the computation of the commutation reactance). Therefore, the model is setup with a commutation reactance $X_c = 4.536\Omega$ for each converter, under the assumption of identical converter transformers for both the recitifer and inverter sides and having a positive-sequence leakage reactance of 12%. [1]

The relevant functions of the *HVDC LCC Monopole CDC4T* model are:

- Load Flow model - regulation of active power flow through the DC link
- Dynamic Model - regulation of active power/current flow through the DC link
- Dynamic Model - controlled behaviour during AC faults
- Dynamic Model - detailed representation of rectifier/inverter firing angle control
- Dynamic Model - Rectifier controls: constant current and minimum firing angle (α_{min})
- Dynamic Model - Inverter controls: voltage, minimum extinction angle (γ_{min}) and constant current control

The model is designed for use in the following analysis types:

- Load Flow calculation,
- Stability (RMS) simulation,
- Electro-Magnetic Transient (EMT) simulation

Hence, after model configuration (refer to Section 4), the above mentioned study types can be executed without any further model modifications/customisations. For using the model with any other calculation functions, appropriate considerations must be taken in order to obtain plausible behaviour. The model equations and control diagrams are open and available for inspection, hence the user can apply further modifications to them, if so required.

[1] Commutation reactance can be calculated as follows: $X_c = \frac{(U_r \cdot turns_ratio)^2}{S_r} \cdot X_t$, where U_r is the nominal voltage at each station (HV side) expressed in kV, S_r is the rated power of the converter station (assumed to be equal with the rated power of the converter transformer) expressed in MVA, $turns_ratio$ is the transformer's primary-to-secondary transformation ratio, and X_t is the transformer leakage reactance expressed in p.u.

3.1 Model Usage Guidelines

3.1.1 Load Flow Analysis

In load flow calculation, the HVDC-LCC element (*ElmHvdclcc*) has the purpose of controlling both the rectifier and the inverter stations such that various load flow objectives are met. For a comprehensive description of the load flow control modes of the *ElmHVdclcc* model refer to its Technical Reference documentation. The rectifier is by default set up to regulate dc power at the rectifier station. The inverter is set by default to regulate dc voltage at the inverter station. This operation mode is regarded as steady state, normal operation mode.

3.1.2 Dynamic RMS-Simulation

The model is adequate for RMS-simulation studies (balanced/unbalanced) considering the following simulation settings:

- Maximum time step: 0.01 s (10 ms)
- Minimum time step: 0.001 s (1 ms)

Note: The above recommends that the simulation time step settings should be within the afore mentioned range. There is no known restriction in terms of the fixed/variable time step option, i.e. both fixed and variable simulation step size adaptation settings are possible. Nevertheless, the user is advised to use the afore mentioned recommended minimum time step, due to the existence of fast reacting HVDC controllers.

Note: The model operates on positive sequence quantities (feedback signals as well as set-points). The model can be used within an unbalanced RMS simulation, nevertheless, the default operation will remain, surely, as a positive sequence control scheme.

3.1.3 Dynamic EMT-Simulation

The model is adequate for EMT-simulation studies considering the following simulation settings:

- Simulation time step: 0.0001 s (0.1 ms)

Note: When using automatic time step adaptation option, ensure that the maximum time step is selected not greater than 0.0001 s.

---

4 Model Configuration

This chapter contains information about the configuration of the HVDC model.

4.1 HVDC-LCC element

In the following section, the general configuration of the HVDC model.

4.1.1 Adjusting Converter Parameters

The HVDC model ratings are available via several *PowerFactory* objects:

- The HVDC-LCC element *ElmHvdclcc*
- The dynamic control DSL common model *HVDC Dynamic Controller*
- The Shunt elements
- The two AC busbars

Adjusting the rated AC power

The rated AC power can be adjusted individually for each converter station. This is done via the HVDC-LCC element *ElmHvdclcc*, “Basic Data”, “Type” reference, “General” tab.

Adjusting other converter ratings

Similar to the change of the rated AC power described above, many other converter ratings may be changed within the HVDC-LCC type object (*TypHvdclcc*) using the “General”, “Converter 1” and “Converter 2” tabs.

Adjusting the rated dc current and voltage

The rated dc current and voltage may be adjusted within the dynamic control model in order to per unitize the dynamic control model. As such, the DSL model *HVDC Dynamic Controller* contains the parameters *IDCRATED* and *VDCRATED*, expressed in kA and kV respectively. These parameters may be adapted as found appropriate.

4.2 Dynamic Model Configuration

The dynamic simulation is based in *PowerFactory* on the load flow calculation. This means that the starting point for the dynamic simulation will always be the load flow solution. It is therefore important to check that the load flow configuration of all calculation relevant elements is correct.

The composite model which contains the links to all dynamic model parts (DSL model, measurement devices, etc) can be accessed via the Data/Network Model Manager.

- Via Data or Network Model Manager: Search the corresponding composite model, right click and select “Edit”.

- After opening the composite model dialog the different components of the HVDC dynamic model can be accessed via a double click in the second column.

Note: The models which are available in *PowerFactory* as templates are configured with a set of default parameters. Whenever necessary, the user is encouraged to adjust all relevant parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the dynamic model) to represent a specific unit.

The dynamic model can be configured via input parameters. These parameters are displayed after opening the dialog of the DSL model *HVDC Dynamic Controller* on the “General” page. The dynamic DSL control model *HVDC Dynamic Controller* contains array parameters as well (i.e. a list of parameters). The arrays can be found on the second tab of the edit dialog, tab named “Advanced 1”.

The parameters listed in Figure 4.1 are specific to the dynamic DSL model. The table provides a parameter mapping between PSS/E data and the *PowerFactory* dynamic DSL control model *HVDC Dynamic Controller* data. Further parameters from PSS/E may need to be used to update/configure the load flow model data of the other composing elements, as described in Section 4.1.1. A number of dynamic model parameters are unique to the *PowerFactory* model and do not have a correspondent to the PSS/E CD4T model. In general, these parameters define the fast inner loop rectifier and inverter controls.

PSSE file	PSSE parameter name	PowerFactory parameter name	Unit	Description
dyr	TIDC	TIDC	[s]	dc current transducer time constant
dyr	TVDC	TVDC	[s]	dc voltage transducer time constant
-	-	IDCRATED	[kA]	Rated DC current
-	-	VDCRATED	[kV]	Rated DC voltage
raw	MDC	MDC	[0/1/2]	Control mode; 0=blocked;1=Power;2=Current
dyr	RSVOLT	RSVOLT	[kV]	minimum dc voltage following blocking
raw	RDC	RDC	[Ohm]	dc line resistance
dyr	RSCUR	RSCUR	[A]	minimum dc current following blocking
raw	RCOMP	RCOMP	[Ohm]	Compensating resistance for voltage control
dyr	V1	V1	[kV]	VDCL Voltage point 1
dyr	V2	V2	[kV]	VDCL Voltage point 2
dyr	V3	V3	[kV]	VDCL Voltage point 3
dyr	C1	C1	[A]	VDCL Current point 1
dyr	C2	C2	[A]	VDCL Current point 2
dyr	C3	C3	[A]	VDCL Current point 3
-	-	DGS_GAMMARGIN		
dyr	C0	C0	[A]	minimum current demand
dyr	TBYPAS	TBYPAS	[s]	minimum bypassing time
dyr	VBLOCK	VBLOCK	[pu]	rectifier AC blocking voltage
dyr	VBYPAS	VBYPAS	[pu]	inverter DC bypassing voltage
dyr	VUNBL	VUNBL	[pu]	rectifier AC unblocking voltage
dyr	VUNBY	VUNBY	[pu]	inverter AC unbypassing voltage
dyr	TBLOCK	TBLOCK	[s]	minimum blocking time
raw	VCMOD	VCMODE	[kV]	Switched mode voltage threshold
dyr	TCMOD	TCMODE	[s]	min. time stays in switched mode
dyr	ALFDY	ALFDY	[deg]	min. alpha for dynamics
raw	DELTI	DELTI	[p.u.]	Inverter current reference margin
-	-	DGS_GUD		
dyr	GAMDY	GAMDY	[deg]	min. gamma for dynamics
-	-	DGS_GGAM		
-	-	DGS_TUD		
-	-	DGS_TGAM		
-	-	DGS_TGPG		
-	-	DGS_GID		
-	-	DGS_TID		
-	-	DGS_IMARG_REC		
-	-	DGS_TUAC		
-	-	DGS_TGAMMIN		
-	-	DGS_ALPH_GRD_LOW		
dyr	VRAMP	VRAMP	[pu/sec]	voltage recovery rate
dyr	CRAMP	CRAMP	[pu/sec]	current recovery rate

Figure 4.1: HVDC-LCC Parameter Mapping with PSS/E's CDC4T Model

5 References

[1] INC. Power Technologies. *PSS/E 35. VOLUME II: PROGRAM APPLICATION GUIDE.*

List of Figures

3.1 HVDC-LCC Single Line Diagram 2

3.2 HVDC-LCC Composite Frame Diagram 3

3.3 HVDC-LCC Control Diagram 4

4.1 HVDC-LCC Parameter Mapping with PSS/E's CDC4T Model 9